FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

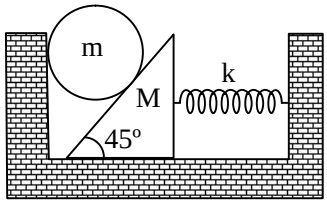
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A hollow vertical cylinder of radius R and height 'h' has smooth internal surface. A small particle is placed in contact with the inner side of the upper rim at a point P. It is given a horizontal speed v_0 tangential to rim. It leaves the lower rim at point Q, vertically below P. The number of revolutions made by the particle will

(A) $\frac{h}{2\pi R}$ (B) $\frac{v_0}{\sqrt{2gh}}$

(C) $\frac{2\pi R}{h}$ (D) $\frac{v_0}{2\pi R}\left(\sqrt{\frac{2h}{g}}\right)$

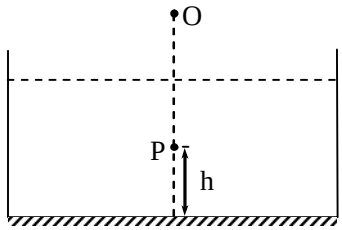
2. All surfaces shown in figure are smooth. System is released with the spring unstretched. In equilibrium, compression in the spring will be



(A) $\frac{2mg}{k}$ (B) $\frac{(M+m)g}{\sqrt{2}k}$

(C) $\frac{mg}{\sqrt{2}k}$ (D) $\frac{mg}{k}$

3. A plane mirror in placed at bottom of a tank containing a liquid of refractive index μ. P is a small object at a height 'h' above the mirror. An observer "O" vertically above P outside the liquid observes P & its image in the mirror. The apparent distance between these two will be

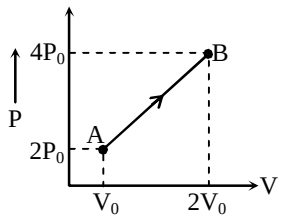


(A) $2\mu h$ (B) $\frac{2h}{\mu}$

(C) $\frac{2h}{\mu-1}$ (D) $h\left(1+\frac{1}{\mu}\right)$

Space for Rough work

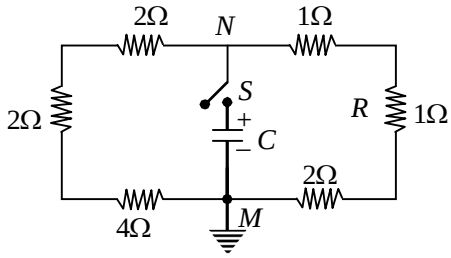
4. One mole of a monoatomic gas undergoes the process A $\rightarrow$ B in the given P-V diagram. The specific heat for this process is



(A) $\frac{3}{2}$R (B) 2R

(C) $\frac{30}{7}$R (D) $\frac{20}{7}$R

5. A capacitor of capacity 6 μF and initial charge 160 μC is connected with a key *S* and resistance as shown in figure. Point *M* is earthed. If key is closed at *t* = 0 then the current through resistance *R* = 1 Ω at *t* = 16 μ sec is



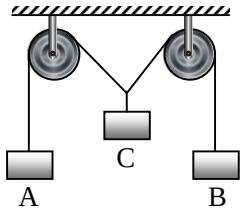
(A) $\frac{10}{3e}A$

(B) $\frac{10}{e}A$

(C) $\frac{20}{3e}A$

(D) None of these

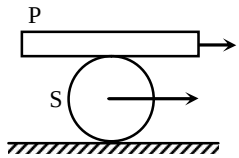
6. Three blocks A, B and C are suspended as shown below. Mass of each of block A and B is *m*. If system is in equilibrium and mass of C is M then



(A) M < 2m (B) M > 2m

(C) M = 2m (D) M $\leq$ 2m

7. A plank P is placed on a solid cylinder S, which rolls on a horizontal surface. The two are of equal mass. There is no slipping at any of the surfaces in contact. The ratio of kinetic energy of P to the kinetic energy of S is



(A) 1 : 1 (B) 2 : 1

(C) 8 : 3 (D) 11 : 8

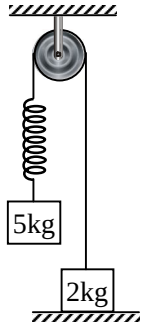
Space for Rough work

8. A particle is projected at an angle of elevation α and after t second it appears to have an angle of elevation β as seen from point of projection. The initial velocity will be

(A) $\frac{gt}{2\sin(\alpha-\beta)}$ (B) $\frac{gt\cos\beta}{2\sin(\alpha-\beta)}$

(C) $\frac{\sin(\alpha-\beta)}{2gt}$ (D) $\frac{2\sin(\alpha-\beta)}{2gt\cos\beta}$

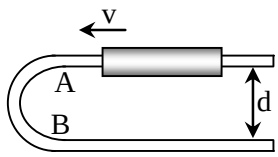
9. System shown in the figure is released from rest with mass 2 kg in contact with the ground. Pulley and spring are massless and the friction is absent everywhere. The speed of 5 kg block when 2 kg block leaves the contact with the ground is (force constant of the spring k = 40 N/m and g = 10 m/s^2)



(A) $\sqrt{2}$ m/s (B) $2\sqrt{2}$ m/s

(C) 2 m/s (D) $\sqrt{2}$ m/s

10. A U-shaped wire has a semicircular bending between A and B as shown. A bead of mass m moving with uniform speed 'v' through a wire enters the semicircular bend at A leaves at B with velocity v/2 after time T. The average force exerted by the bead on the part AB of the wire is



(A) 0 (B) $\frac{3mv}{2T}$

(C) $\frac{3mv}{T}$ (D) None of these

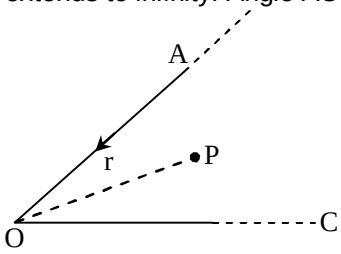
11. There are two thin wire rings, each of radius R, whose axes coincide. The charges on the rings are q and –q. Evaluate the potential difference between the centres of the rings separated by a distance *a*.

(A) $\frac{q}{\pi\varepsilon_0}\left[\frac{1}{R}+\frac{1}{\sqrt{R^2+a^2}}\right]$ (B) $\frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R}+\frac{1}{\sqrt{R^2+a^2}}\right]$

(C) $\frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+a^2}}\right]$ (D) $\frac{2q}{\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+a^2}}\right]$

Space for Rough work

12. Two wires AO and OC carry equal currents i as shown in figure. One end of both the wires extends to infinity. Angle AOC is α



The magnitude of magnetic field at a point P on the bisector of these two wires at a distance *r* from point O is

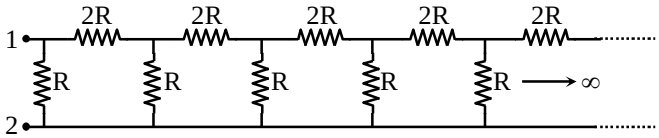
(A) $\frac{\mu_0}{2\pi}\frac{i}{r}\cot\left(\frac{\alpha}{2}\right)$

(B) $\frac{\mu_0}{4\pi}\frac{i}{r}\cot\left(\frac{\alpha}{2}\right)$

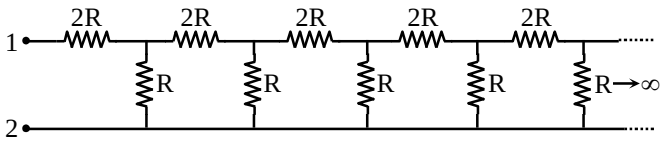
(C) $\frac{\mu_0}{2\pi}\frac{i}{r}\frac{\left(1+\cos\frac{\alpha}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)}$

(D) $\frac{\mu_0}{4\pi}\frac{i}{r}\left(\frac{\alpha}{2}\right)$

13. Two circuits (as shown in figure) are called Circuit A and Circuit B. The equivalent resistance of Circuit A is *x* and that of Circuit B is *y* between 1 and 2



Circuit A



Circuit B

(A) y = x

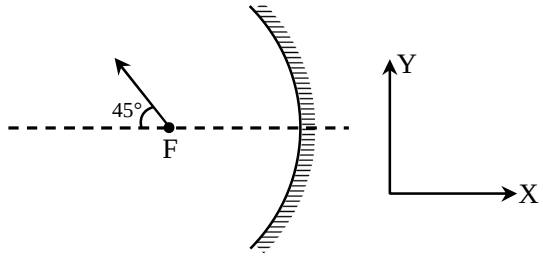
(B) $y = (\sqrt{3}+2)R$

(C) $xy = 2R^2$

(D) x + y = 2R

Space for Rough work

14. A particle is projected on a horizontal (x-z plane) at an angle 45° with horizontal as shown. The particle is projected from focus F of a concave mirror of radius of curvature R with speed $v = \sqrt{gR}$. Velocity of image of the particle as it is just about to strike the horizontal plane is

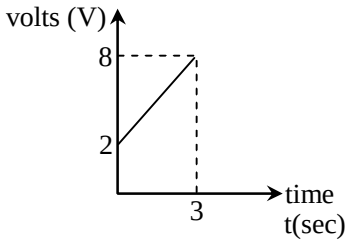


(A) $\frac{v}{4\sqrt{2}}\hat{i} - \frac{v}{4\sqrt{2}}\hat{j}$ (B) $\frac{v}{4\sqrt{2}}\hat{i} + \frac{v}{2\sqrt{2}}\hat{j}$

(C) $\frac{v}{4\sqrt{2}}\hat{i} - \frac{v}{2\sqrt{2}}\hat{j}$ (D) $\frac{-v}{4\sqrt{2}}\hat{i} + \frac{v}{4\sqrt{2}}\hat{j}$

15. A string of length 1.5 m with its two ends clamped is vibrating in fundamental mode. Amplitude at the centre of string in 4 mm. Distance two point having amplitude 2 mm is
(A) 1 m (B) 75 cm
(C) 60 cm (D) 50 cm

16. A circuit element is placed in a closed box. A time t = 0, a constant current generator supplying a current of I amp is connected across the box. Potential difference across the box varies according to graph as shown in the figure. The element in the box is



(A) a resistance of 2Ω
(B) a battery of emf 6V
(C) an inductance of 2H
(D) a capacitance

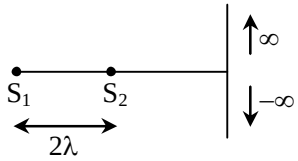
17. A charged particle of specific charge α mass with a velocity $\vec{v} = v_0\hat{i}$ in a magnetic field $\vec{B} = \frac{B_0}{\sqrt{2}}(\hat{j} + \hat{k})$, then
(A) Path of the particle is a helix
(B) Path of the particle is a ellipse
(C) Distance moved by the particle in time $t = \frac{\pi}{B_0\alpha}$ is $\frac{\pi v_0}{B_0\alpha}$
(D) Velocity of particle after the time t = $\frac{\pi}{B_0\alpha}$ is $\left(\frac{v_0}{2}\hat{i} + \frac{v_0}{2}\hat{j}\right)$

Space for Rough work

18. The shape of a wave propagating in positive X-direction or negative x-direction is given by, $y = \frac{1}{\sqrt{1+x^2}}$ at t = 0 and $y = \frac{1}{\sqrt{2-2x+x^2}}$ at t = 1 sec where x & y are in metres. The shape of wave disturbance does out hange during propagation. The velocity of wave is
(A) 1 m/sec in positive X-direction (B) 1 m/sec in negative X-direction
(C) $\frac{1}{2}$ m/sec in positive X-direction (D) $\frac{1}{2}$ m/sec in negative X-direction

19. A steel rod is 4 cm is diameter at 30º. A brass ring has an interior diameter 3.992 cm at 30ºC. In order that the ring just slides onto the steel rod, the common temperature of the two should be ($\alpha_{steel} = 11 \times 10^{-6}$/ºC & $\alpha_{brass} = 19 \times 10^{-6}$/ºC)
(A) 200ºC (B) 350ºC
(C) 280ºC (D) 300ºC

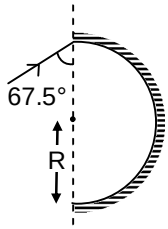
20. There are two source kept at distance of 2λ. A large screen is perpendicular to line joining the sources. Number of maximas on the screen in this case is (λ wavelength of light) [Excluding at ∞ and $-\infty$]
(A) 1 (B) 3
(C) 5 (D) 7



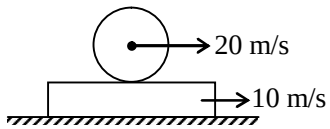
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A Ray incident on a spherical mirror which subtends an angle θ on its center of curvature where θ is slightly greater than 180°. Find number of reflection that may occurs before the ray comes out of the mirror. (including the first reflection)



22. A solid cylinder is rolling without slipping on a plank which also moving with speed 10 m/s on a horizontal surface as shown. The speed of centre of cylinder is 20 m/s. The mass of cylinder is 2 kg. The kinetic energy of the cylinder when observed from ground is 90 nJ. Then find value of n.

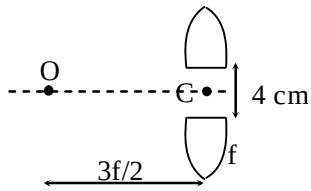


Space for Rough work

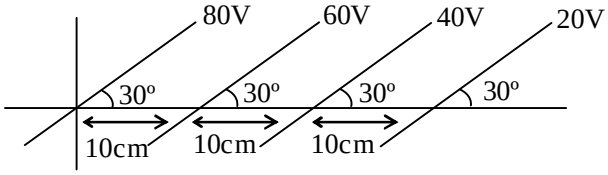
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A split lens has two parts separated by 4 cm and focal length is f. An object is placed at a distance 3f/2 from C as shown in figure. The distance between the images formed by the two halves (in cm) is.



24. In a space, equipotential surfaces are shown below.



Then electric field in the space is

25. In an α - decay, the kinetic energy of α -particle is 48 MeV and Q value of the reaction is 50 MeV. The mass number of the mother nucleus will be (assume that daughter nucleus is in ground state).

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. A beer has pH of 4.3. What is the $[H_3O^+]$?
(A) 3×10^{-4} (B) 2×10^{-4}
(C) 3×10^{-5} (D) 5×10^{-5}

27. In the given plot of lnKeq versus 1/T: (T = temperature) for a reaction is shown below the reaction must be
(A) exothermic (B) endothermic
(C) one with negligible enthalpy change (D) highly spontaneous at ordinary temperature

28. Which gas shows real behaviour?
(A) 16 g O_2 at 1 atm and 273 K occupies 11.2 L
(B) 1 g H_2 in 0.5 L flask exerts pressure of 24.63 atm at 300 K
(C) 1 mole NH_3 at 300 K and 1 atm occupies volume 22.4 L
(D) 5.6 of CO_2 at 1 atm and 273 K is equal to 11 gm

29. Which set of quantum numbers is impossible for an electron?
(A) n = 1, l = 0, ml = 0, ms = +1/2 (B) n = 9, l = 7, ml = –6, m_s = –1/2
(C) n = 2, l = 1, ml = 0, m_s = +1/2 (D) n = 3, l = 2, ml = –3, m_s = +1/2

30. Balmer gave an equation for wavelength for visible region of H – spectrum as $\lambda = \frac{Kn^2}{n^2 - 4}$
(n = principal quantum no.) then 'K' in terms of 'R'. (R = Rydberg constant)
(A) R (B) R/2
(C) 4/R (D) 5/R

31. Cadmium amalgam is prepared by electrolysis of a solution of $CdCl_2$ using a mercury cathode. How long should a current of "4A" be passed in order to prepare 10% by mass Cd in Cd – Hg amalgam on cathode of 4.5 g of Hg? (atomic mass of Cd = 112)
(A) 400 sec (B) 215.40 sec
(C) 861.6 sec (D) 430.8 sec

Space for Rough work

32. The value of $\frac{T_{0.875}}{T_{0.5}}$ for nth order of reaction is

(A) $2^{(2n-2)}$ (B) $2^{2(n-1)-1}$

(C) $\frac{8^{n-1}-1}{2^{n-1}-1}$ (D) none of these

33. When two mole of an ideal gas $\left(C_{p,m}=\frac{5}{2}R\right)$ heated from 300 K to at constant pressure, the change in entropy of gas (Δs) is

(A) $\frac{3}{2}R\ln 2$ (B) $-\frac{3}{2}(\ln 2)R$

(C) $5R\ln 2$ (D) $\frac{5}{2}R\ln 2$

34. Which of the following layering pattern will have a void fraction of 0.26?

(A) ABCCBAABC (B) ABBABBA

(C) ABCABCABC (D) ABCAABCA

35. The $"O-\hat{N}-O"$ bond angle is maximum in

(A) NO_2 (B) NO_2^+

(C) NO_2^- (D) NO_3^-

36. Which of the following are isoelectronic and isostructural?

(A) NO_3^-, CO_3^{2-} (B) SO_3, NO_3^-

(C) ClO_3^-, CO_3^{2-} (D) CO_3^{2-}, ClO_3^-

37. Which of the following statement is not correct

(A) bis (glycinato) zinc (II) is optically active

(B) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$ have different shape

(C) $[Ni(CN)_4]^{4-}$ is square planar complex

(D) $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ have the same magnetic moment

38. In the alumino – thermite process, Al metal act as

(A) oxidizing agent (B) reducting agent

(C) catalyst (D) flux

Space for Rough work

39. Which of the given complex species is following E.A.N rule?

(A) $\left[Ca(EDTA)\right]^{2-}$ (B) $\left[Cr(en)_3\right]^{3+}$

(C) $K_3\left[COBr_2Cl_2(NO_3)_2\right]$ (D) $[Ni(dmg)_2]$

40. BHC and DDT act as

(A) carcinogens (B) allergens

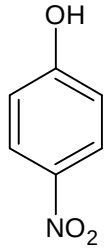
(C) asthmatic agents (D) all of these

41. For this compound $\overset{+}{N}(Me)_3 - CH_2CH_2I$, which form is most stable?

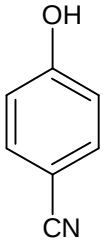
(A) anti form about C – C (B) gauche form about C – C

(C) partially eclipsed about C – C (D) fully eclipsed about C – C

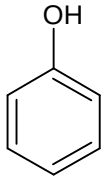
42. Correct order of acidic strength for the following compounds



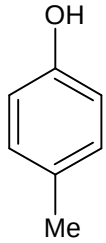
(i)



(ii)



(iii)

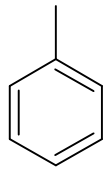


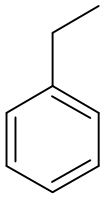
(iv)

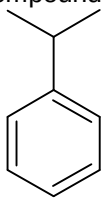
(A) i < ii < iii < iv (B) ii > i > iii > iv

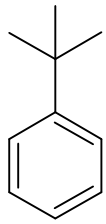
(C) i > ii > iv > iii (D) i > ii > iii > iv

43. Correct order of stability for the following compound


(i)


(ii)

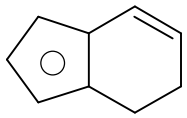

(iii)


(iv)

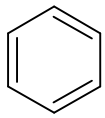
(A) i > ii > iii > iv (B) i < ii < iv < iii

(C) i > ii > iv > iii (D) i < ii < iii < iv

Space for Rough work

44.  $\xrightarrow{\text{cold dil. } KMnO_4}$ product of the reaction is

(A) meso compound
(B) enantiomeric pair
(C) diastereomers
(D) optically pure enantiomer

45.  $\xrightarrow[\text{Liq.}NH_3]{\text{Na}}$ (A) $\xrightarrow[\text{Zn}]{O_3}$ (B) $\xrightarrow{Ph_3P = CH_3 \text{ (2 mole)}}$ (C)

Product (C) of the above reaction is
(A) 1, 3-hexadiene
(B) 1, 4-pentadiene
(C) 1, 3-butadiene
(D) 1, 3-heptadiene

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. Number of S – S bond in $H_2S_nO_6$ is 7 then what is the value of 'n' …

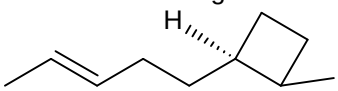
47. 10 ml of N – HCl, 20 ml of N/2 H_2SO_4 and 30 ml of N/3 HNO_3 are mixed together and volume made to 2 litre. The normality of H^+ in the resulting solution 'x' then $\frac{200x}{N} = ?$

Space for Rough work

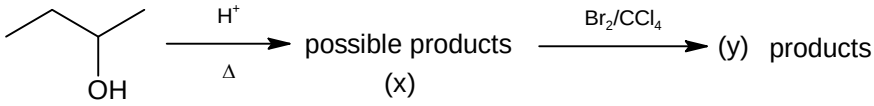
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Total numbers of geometrical isomers of the following compound is x then x =?



49. How many carbon-hydrogen bond orbitals are available for overlap with the vacant p-orbital in ethylcarbocation?

50. 

The number of possible products for x and y is z and w then "z + w" equal to …

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. $3\tan^6\left(\frac{\pi}{18}\right) - 27\tan^4\left(\frac{\pi}{18}\right) + 33\tan^2\left(\frac{\pi}{18}\right)$ equals.

(A) 0 (B) 1
(C) 2 (D) 3

52. Consider the cubic equation
$x^3 - (1+\cos\theta+\sin\theta)x^2 + (\cos\theta\sin\theta+\cos\theta+\sin\theta)x - \sin\theta\cos\theta = 0$ where x_1, x_2, x_3 are roots of cubic.
Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal.

(A) 3 (B) 4
(C) 5 (D) 6

53. The number of positive integral solution of $x^4 - y^4 = 3789108$

(A) 0 (B) 1
(C) 2 (D) 4

54. If p, q, r are the substatement with truth values, T, T, F then the statement.
$r \to (p \wedge \sim q) \vee (\sim q \wedge \sim r)$ will be

(A) True (B) False
(C) May be true or false (D) None of these

55. A void relation on set A is

(A) Reflexive (B) Symmetric and Transitive
(C) Reflexive and Symmetric (D) Reflexive and Transitive

Space for Rough work

56. All values of 'θ' between 0 and π which satisfy the equation $\cos\theta\cos 2\theta\cos 3\theta = \frac{1}{4}$ is

(A) $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$ (B) $\frac{\pi}{8}, \frac{\pi}{3}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

(C) $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$ (D) None of these

57. Let $f: R \to R$ and $g: R \to R$ be two one – one onto functions such that they are mirror image of each other about y = 0 then $h(x) = f(x) + g(x)$ is:-

(A) One – One and onto (B) One – One but not, onto
(C) Not one one but onto (D) Neither one – one nor onto

58. If $\sum_{i=1}^{2n} \sin^{-1} x_i = n\pi$ then $\sum_{i=1}^{2n} x_i$ is

(A) n (B) 2n
(C) 4n (D) 0

59. $\lim_{n\to\infty} \frac{n^k \cos n!}{n+1}; (0 < k < 1) =$

(A) 0 (B) 1!
(C) 2! (D) None of these

60. Let $f(x) = \sqrt{\sin^{-1} x} + \sqrt{\cos^{-1} x}$ defined, $In[0,1]$ then

(A) $f(x-3)$ will be continuous, $In[3,4]$ (B) $f(x-3)$ will be continuous, $In[0,1]$
(C) $f(x-3)$ will be continuous, $In[0,1)$ (D) $f(x-3)$ will be continuous, $In(0,1)$

61. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$ then $\frac{dy}{dx}$ is

(A) $\frac{x^2}{y^2}\sqrt{\frac{1-x^6}{1-y^6}}$ (B) $\frac{y^2}{x^2}\sqrt{\frac{1-y^6}{1-x^6}}$

(C) $\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$ (D) $\frac{y^2}{x^2}\sqrt{\frac{1-x^6}{1-y^6}}$

Space for Rough work

62. Let f(x) = $1+x^m(x-1)^n$ where m, n then In (0, 1) the equation $f'(x)=0$ has
(A) No root
(B) At least one root
(C) At most one root
(D) Exactly one

63. For the curve $\frac{c^4}{r^2}=\frac{a^2}{\sin^2\theta}+\frac{b^2}{\cos^2\theta}$, the maximum value of r is
(A) $\frac{c^2}{a+b}$
(B) $\frac{a+b}{c^2}$
(C) $\frac{c^2}{a-b}$
(D) $c^2(a+b)$

64. Let $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $N=\prod_{n=1}^{\infty}\frac{\left(1+n^{-1}\right)^2}{\left(1+2n^{-1}\right)}$ then value of $L^{-1}+M^{-1}+N^{-1}$
(A) 6
(B) 8
(C) 10
(D) 12

65. $\sin\frac{\pi}{900}\left\{\sum_{r=1}^{10}(r-\omega)\left(r-\omega^2\right)\right\}=\{\text{where '}\omega\text{' is cube root of unity}\}$
(A) 0
(B) $\frac{1}{2}$
(C) 1
(D) $\frac{1}{\sqrt{2}}$

66. Let A, B, C are three mutually independent events, consider two statements S_1 and S_2
S_1: A and $B\cup C$ independent
S_2: A and $B\cap C$ independent
Then
(A) Both S_1 and S_2 are true
(B) Only S_1 is true
(C) Only S_2 is true
(D) Both S_1 and S_2 are false

Space for Rough work

67. A bag constants 12 pair of socks, 4 socks are drawn of random. The probability that there is at least one pair is

(A) $\frac{7}{25}$ (B) $\frac{40}{161}$

(C) $\frac{41}{161}$ (D) $\frac{8}{25}$

68. If A is a square matrix of order 3, then correct statement is

(A) det (-A) = - det A (B) det (-A) = 0

(C) det (-A + I) = I + det (A) (D) det (2A) = 2 det A

69. If $\vec{p}, \vec{q}, \vec{r}$ be three mutually perpendicular vectors of same magnitude. If a vector $\vec{x}$ staties the equation

$\vec{p}\times\left(\left(\vec{x}-\vec{q}\right)\times\vec{p}\right)+\vec{q}\times\left(\left(\vec{x}-\vec{r}\right)\times\vec{q}\right)+\vec{r}\times\left(\left(\vec{x}-\vec{p}\right)\times\vec{r}\right)=0$ then $\vec{x}$ is given by

(A) $\frac{1}{2}\left(\vec{p}\times\vec{q}\times-2\vec{r}\right)$ (B) $\frac{1}{2}\left(\vec{p}+\vec{q}+\vec{r}\right)$

(C) $\frac{1}{3}\left(\vec{p}+\vec{q}+\vec{r}\right)$ (D) $\frac{1}{3}\left(2\vec{p}+\vec{q}-\vec{r}\right)$

70. The value of $\int\frac{dt}{t^2+2xt+1}\left(x^2>1\right)$is

(A) $\frac{1}{\sqrt{1-x^2}}\tan^{-1}\left(\frac{t+x}{\sqrt{1+x^2}}\right)+C$ (B) $\frac{1}{2\sqrt{x^2-1}}\log\left(\frac{t+x-\sqrt{x^2-1}}{t+x+\sqrt{x^2-1}}\right)+C$

(C) $\frac{1}{2}\log\left(t^2+2xt\right)+C$ (D) None of these

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The reflection of curve xy = 1 the line y = 2x is the curve $12x^2 - rxy + 5y^2 + t = 0$ then value of r is

72. The least value of $2\log_{100} a - \log_a 0.0001, a > 1$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. If $L = \lim_{x\to 0} \frac{ae^x - \cos 2x + \sin bx}{x^2}$ exists and is infinite then value of L is

74. Number of ways of arranging the letters of word ALGEBRA in which either vowels or constant, but not both appear in alphabetical order is.

75. In a right angled isosceles triangle, orthocenter is (2, –3) and circumcenter is 9–1, 1). The radius of the circle scribed in it is. (Take $\sqrt{2} = 1.414$)

Space for Rough work